

Binary outcomes, logit, probit, and likelihood

Slide route: Lecture11v2.tex: LPM, nonlinear models, effects, MLE and model selection. Chapter 11. Core route plus assigned extensions.

Session 15 | Model a probability

Outcomes: Interpret an LPM coefficient, calculate a logit probability and marginal effect, explain likelihood, and distinguish odds from probabilities.

100-minute core route

0–10	Why a binary outcome has a conditional probability as its mean.
10–25	LPM and its properties.
25–45	Logit/probit links and predicted probabilities.
45–55	Pause and coefficient interpretation.
55–75	Marginal effects, discrete changes, odds.
75–90	Bernoulli likelihood and estimation intuition.
90–100	One application table and exit ticket.

This is a dense deck. Assign repeated coefficient screenshots, the extended MLE algebra, and detailed model-selection slides for reading. Revisit AIC/BIC and model comparison in session 16. Do not try to narrate every slide.

Slide-by-slide script

Slide 1 | Title

Guide Open with the practical question: how do we model an outcome that is only zero or one?

Slide 2 | Roadmap

Guide Preview LPM, nonlinear links, marginal effects, likelihood, and model selection.

Slide 3 | Last lesson

Guide Retrieve a fixed-effects or clustering assumption and explain why binary outcomes require a new conditional-mean model.

Slide 4 | Binary choice models

Guide State $\mathbb{E}(Y | X) = P(Y = 1 | X)$ when Y is binary. This is the bridge to probability models.

Slide 5 | Binary choice models

Guide Ask students to identify the unit, binary outcome, and regressors in the application.

Slide 6 | Linear probability model

Guide Write $p(X) = \beta_0 + \beta_1 X$. Interpret β_1 as a probability-point change per unit of X .

Slide 7 | Linear probability model

Guide Use the synthetic $\hat{p} = .2 + .1X$ and calculate the prediction at $X = 9$.

Slide 8 | LPM properties

Guide Explain why fitted probabilities can leave $[0, 1]$ and why binary errors are heteroskedastic.

Slide 9 | LPM properties

Guide Ask whether robust SEs fix the range problem. No; they only address a variance calculation.

Slide 10 | LPM

Guide Discuss interpretation and the role of the conditional mean. Keep the percentage-point units explicit.

Slide 11 | LPM

Guide Compare a predicted probability with an observed zero or one; prediction is a mean, not an individual certainty.

Slide 12 | Nonlinear models

Guide Motivate a link function that maps every index to a probability in $[0, 1]$.

Slide 13 | Nonlinear models

Guide Draw the S-shaped curve and ask where the marginal effect is largest.

Slide 14 | Probit model

Guide Define $p(X) = \Phi(X'\beta)$ and distinguish the latent index from the probability.

Slide 15 | Probit model

Guide Calculate a probability from a simple index if the slide gives one; focus on the mapping, not table lookup mechanics.

Slide 16 | Logit model

Guide Write $\Lambda(z) = 1/(1 + e^{-z})$ and calculate $\Lambda(.5) \simeq .6225$.

Slide 17 | Probit interpretation

Guide Explain that a probit coefficient shifts the latent index; it is not directly a percentage-point effect.

Slide 18 | Probit interpretation

Guide Use the derivative $\phi(X'\beta)\beta_1$ to show why effects vary with X .

Slide 19 | Probit interpretation

Guide Ask students to state the value of the other covariates whenever they report a marginal effect.

Slide 20 | Probit interpretation

Guide Contrast a derivative for a continuous regressor with a discrete change for a dummy.

Slide 21 | Logit interpretation

Guide State that raw logit and probit coefficients are on different index scales; compare probabilities or marginal effects.

Slide 22 | Logit interpretation

Guide Use the same index idea to explain the sign of a coefficient without overinterpreting its magnitude.

Slide 23 | Marginal effects

Guide Calculate the derivative of the logit probability, $\Lambda(z)[1 - \Lambda(z)]\beta_1$.

Slide 24 | Marginal effects

Guide Show why the marginal effect is largest around probability .5 and smaller near zero or one.

Slide 25 | Marginal effects

Guide For a binary regressor, calculate $p(D = 1) - p(D = 0)$ rather than a derivative.

Slide 26 | Marginal effects

Guide Distinguish an effect at the mean covariates from the average of individual effects.

Slide 27 | Marginal effects

Guide Ask students to interpret a positive marginal effect in percentage points, not percent unless the outcome is logged.

Slide 28 | Marginal effects, logit

Guide Use $X = 4$ and $X = 5$ in the synthetic model: .5 and .6225, a 12.25-point change.

Slide 29 | Marginal effects, logit

Guide Connect the numerical change to the curve and ask why the same coefficient can imply different probability changes.

Slide 30 | Probit or logit?

Guide Say that substantive conclusions often agree; compare fitted probabilities, marginal effects, and diagnostics.

Slide 31 | Fitted probabilities

Guide Check that nonlinear fitted values lie in $[0, 1]$.

Slide 32 | Log-odds ratio

Guide Define odds $p/(1 - p)$ and log odds. Do not call log odds a probability.

Slide 33 | Log-odds units

Guide Use a small coefficient to show that an odds ratio is e^β , not β percentage points.

Slide 34 | Maximum likelihood

Guide Explain that the estimator chooses parameters making the observed outcomes likely under the model.

Slide 35 | Estimation and inference

Guide Separate the objective function from the standard errors used for inference.

Slide 36 | Estimation and inference

Guide Ask what a large-sample z statistic is testing in a logit or probit model.

Slide 37 | Bernoulli example

Guide Write the likelihood contribution $p_i^{y_i}(1 - p_i)^{1-y_i}$ for one observation.

Slide 38 | Bernoulli example

Guide Multiply contributions across independent observations and use logs to turn products into sums.

Slide 39 | Bernoulli example

Guide Calculate a tiny two-observation likelihood to make the product concrete.

Slide 40 | Maximum likelihood

Guide Explain why the score is set to zero at an interior maximum and why numerical optimization is common.

Slide 41 | Maximum likelihood

Guide Mention separation: no finite logistic coefficient may exist when groups are perfectly classified.

Slide 42 | Model selection criteria

Guide Define AIC and BIC as fit measures penalized for parameters; compare only compatible samples and likelihoods.

Slide 43 | Model selection criteria

Guide Use the class-16 extension with $\ell_A = -100$, $k_A = 3$ and $\ell_B = -96$, $k_B = 5$, $n = 200$.

Slide 44 | Model selection criteria

Guide Calculate $AIC = 2k - 2\ell$ and $BIC = \log(n)k - 2\ell$ for both models.

Slide 45 | Application

Guide Name the labor-force participation outcome and the key covariates before reading the table.

Slide 46 | Application: LPM

Guide Interpret the coefficient in probability points and note the range limitation.

Slide 47 | Application: Probit

Guide Interpret the sign and use marginal effects for probability units.

Slide 48 | Application: Logit

Guide Use odds ratios or marginal effects, not raw coefficients as percentage points.

Slide 49 | Application comparison

Guide Ask whether model choice changes the causal assumptions. It does not solve selection or omitted variables.

Slide 50 | Summary

Guide Have students distinguish probability, odds, marginal effect, and likelihood in four short answers.

Slide 51 | Closing

Guide Set up the cumulative review: choose a model for a question and defend the assumptions before calculating.

Worked example | LPM versus logit

Use with slides Lecture11v2 slides 4–16: Binary Choice Models, Linear Probability Model, Properties of the LPM, Nonlinear Models, and Logit Model. Use the LPM range problem on slides 6–11 and the logit probability calculation on slides 12–16.

For a synthetic LPM $\hat{p} = .2 + .1X$, raising X by one gives a ten-percentage-point change. At $X = 9$, it predicts 1.1, illustrating the range problem. Binary errors have conditional variance $p(X)[1 - p(X)]$, so heteroskedasticity is natural.

$$p(X) = \Lambda(-2 + .5X) = \frac{1}{1 + \exp(2 - .5X)}.$$

At $X = 4$, $p = .5$. At $X = 5$, $p = \Lambda(.5) \simeq .6225$. The one-unit discrete change is .1225, or 12.25 percentage points.

$$\frac{\partial p}{\partial X} = .5p(1 - p); \quad \text{at } X = 4, \quad .5(.5)(.5) = .125.$$

The derivative and finite change are close, not identical.

Interpretation, probit, and likelihood

Use with slides Lecture11v2 slides 17–30 for probit/logit coefficient interpretation, marginal effects, and odds; slides 31–39 for MLE and the Bernoulli likelihood.

For the logit example, a one-unit X increase multiplies the odds by $e^{.5} \simeq 1.649$. This is not a 64.9-percentage-point rise in probability.

Probit If $p(X) = \Phi(\beta_0 + \beta_1 X)$, its derivative is $\phi(\beta_0 + \beta_1 X)\beta_1$. Both logit and probit coefficients move an index; compare predicted probabilities or marginal effects rather than raw coefficient magnitudes across links.

Binary regressor For a dummy D , report a discrete probability change between $D = 0$ and $D = 1$ at specified other covariates, or average those individual changes. A derivative is not the natural one-unit effect for a binary regressor.

Board example | Bernoulli likelihood

For independent observations $y = (1, 0, 1)$ and a common success probability p :

$$L(p) = p^2(1 - p), \quad \ell(p) = 2 \log p + \log(1 - p).$$

$$\ell'(p) = 2/p - 1/(1 - p) = 0 \quad \Rightarrow \quad \hat{p} = 2/3.$$

For regression, replace the common p with $p_i = F(X_i'\beta)$:

$$\ell(\beta) = \sum_i \{y_i \log p_i + (1 - y_i) \log(1 - p_i)\}.$$

Software maximizes this expression numerically. The objective rewards high probability for observed outcomes; it does not minimize squared residuals.

Practical caution Complete separation can prevent finite logistic MLE coefficients. Tiny standard errors or convergence messages should be read, not ignored.

Exit answers An LPM coefficient .03 means three percentage points per unit. A logit coefficient .03 is not generally three percentage points. Average marginal effects average observation-specific effects; effects at mean covariates are a different calculation.

Preparation For session 16, bring a one-paragraph causal question and propose an estimator with its key assumption. Read the remaining model-selection slides; do not treat classification accuracy as evidence of causality.

Source: Lecture11v2.tex. Numerical examples are synthetic; extended MLE/model selection are assigned preparation for review.